

Synthetic Fusion: Supplementary Methods and Results

Abstract

This supplementary material provides detailed methods, metric definitions, results, and limitations for the main manuscript. It also explains the records in the accompanying machine-readable artifact package. The accompanying Chapter 3 reproducibility repository documents the code and required inputs.

S1 Scope and Organization

The main manuscript presents the method, design requirements, quality and candidate-selection decisions, and comparisons with baseline models. This supplement provides the recorded definitions, assignment rules, metrics, thresholds, and results needed to inspect those decisions. It also maps the reported findings to the artifact records and describes the applications to other datasets.

Table S1: Supplementary document map.

Section	Content
S1	Document map and artifact checks of the governance rules.
S2	Role-assignment rules and branch fitting details.
S3	Artifact structure, role ownership, logical columns, and quality filtering.
S4	Support-overlap definition, threshold rationale, and mapping to data-quality concepts.
S5	Artifact package index, audit procedure, checksums, schemas, and temporal stability diagnostics.
S6	Protocol coverage, candidate columns, sparse-window diagnostics, continuous subsets, and baseline definitions.
S7	Procedure for new deployments and detailed Smart* and TON_IoT records.
S8	Extended design requirements and the candidate-filtering figure.
S9	Decision rules, analytical examples, result scopes, and controlled experiment.
S10	Implementation details, feature interpretation, task definitions, and TON_IoT window sensitivity.

The main manuscript defines the terminology and reviews related work in Sections 2–3. The records below provide the additional information needed to inspect the implementation and its outputs.

S1.1 Checks of the Governance Rules

Table S2 links the governance rules in the main manuscript to the artifact records. Reviewers can use these records to check how each rule is represented in the saved outputs.

Table S2: Artifact records for checking the governance rules in the main manuscript. The CSV schema catalog describes the fields in each record.

Governance rule	Artifact record	Reviewer check
Unique ownership	Claim-scope table and role-manifest rows in the package index.	Confirm that each evaluated column has one owner and that excluded columns and placeholders do not support positive quality claims.
Split monotonicity	Candidate-selection ledgers, TRAIN/VAL/TEST manifests, and Smart* gate-freeze audit records.	Check each fitting stage separately: branch development uses TRAIN, selection uses VAL, and the final observation-mask generator uses TRAIN+VAL. TEST exposure limits confirmation. For Smart*, check the policy, thresholds, role map, split, selection, and TEST records using their timestamps, where available, and SHA-256 hashes.
Smart* threshold definitions	Smart* policy and metric records.	Inspect the metric definitions as well as numeric limits. Similar threshold values alone do not establish the same evaluation protocol.
Enforce-before-score	Structural QA ledger and paper-value ledger rows for inactive-finite and role-owner violations.	Confirm that structural violations are checked before the quality metrics are interpreted.
Observed values	Continuous-value and observation-mask records.	The continuous calculations remove nonfinite values separately from each dataset. Q3 separately assesses reporting behavior.
Blocker monotonicity	Q4 blocker ledger, claim-scope table, and release manifest.	Confirm that fatal grades and blocker flags restrict or deny dependent claims. Unrelated passes must not override them.
Scope-compatible baselines	Baseline-fairness comparison ledger.	Confirm that baseline rows report scope, coverage, status, hyperparameters, seeds, and failure/exclusion reasons.
Release-subset monotonicity	Release-scope table and selected-candidate directory.	Confirm that the selected candidate is a documented subset of the scientific artifact selected under Q6.

S2 Governance Procedure and Branch Methods

This section adds the role-resolution rules and branch fitting details to the workflow in main manuscript Section 3.

The workflow is given in Algorithm 1 of the main manuscript. Table S3 specifies the assignment precedence and saved outputs used to implement it.

Table S3: Supplementary role-assignment protocol and audit outputs. The reported queue records assignment checks. Its row count does not establish independent review or agreement.

Precedence step	Rule family	Audit interpretation
Explicit contracts	Declared sparse-driver contracts map to driver_event_target . Canonical time and telemetry_in_sec helpers map to conditioning metadata.	Contracted event drivers and time helpers are frozen before generator fitting.
Reporting-process patterns	Name patterns for obs , present , active , stale , staleness , and mask map to observability or staleness roles before value/state fallback.	Observability is modeled as a CPS process in its own right.
TRAIN/VAL eligibility	Exclude columns without numeric support in TRAIN and VAL before TEST scoring or release assessment.	TEST cannot make a column eligible if fitting and selection lack the required support.
State/value fallback	Binary state-like support maps to binary_state_target . Nonbinary state-like or __value support maps to continuous_value_target . Residual binary/nonbinary numeric columns follow the same support rule.	A numerical battery percentage follows value ownership unless an explicit reporting-process contract takes precedence. A contracted binary event remains a driver.
Assembly exception	A named binary-valued sensor exception is admitted by the binary resolver only after its TRAIN/VAL domain check. At assembly, binary output takes priority over a shared continuous output.	The cups-value column in main manuscript Section 3.3 explains the reduction from 148 source targets to 147 final continuous owners.
Audit outputs	Role-assignment ledgers report 10 precedence rules, 115 automatically flagged ambiguity-queue rows, and 0 duplicate-owner columns.	The initial role map and final assembly registry are different records. Use the assembly registry for final owner counts.

Table S3 groups the ten assignment rules and the separate assembly rule. The 115 flagged fields identify cases where automatic assignment needs closer interpretation. Main manuscript Section 3.3 reports their observed-value categories and semantic labels. Section S10 adds the sampling method and temporal examples. The author review and automatic ownership list answer different questions and should not be treated as an agreement statistic.

Binary admission tolerance. The resolver requires at least one finite TRAIN/VAL value and accepts a field when every finite value lies within 10^{-6} of 0 or 1. Constant observations can satisfy this rule. Main manuscript Section 3.3 explains the milk-cup field’s meaning and assembly ownership, while Section S10 specifies how the source and assembled lists enter the calculations.

Table S4: Branch methods. Candidates use TRAIN for development and VAL for selection. The final observation-mask generator uses TRAIN+VAL. TEST provides the evaluation period.

Branch	Fitting and selection	Output, checks, and limits
Protocol	Estimate supports, count domains, temporal blocks, and candidate families on TRAIN. Select compatible candidates and coupling variants on VAL.	Write TEST-length sequences under nonnegative, integer/count, ownership, and schema constraints. Claims require the relevant Q1/Q2/Q6 evidence and Q4 evidence where paired.
Continuous	Estimate domains, low-cardinality supports, empirical distributions, and candidate generators on TRAIN. Select and calibrate per target on VAL.	Apply the selected observation mask and make inactive positions missing. In the recorded continuous assembly, all selected mask positions are active (Section S10). Enforce domains and support. The 148-target branch does not support a claim that all continuous targets are realistic.
Binary states	Estimate rates, transitions, dwell/run distributions, and cold-start priors on TRAIN. Select using rate, transition, dwell, run-length, and C2ST evidence on VAL.	Enforce binary values and owner rules. Binary overrides are scored in this branch. Scientific claims require qualification. The branch is excluded from quality selection.
Sparse drivers	Estimate event rates, bursts, copy-risk signals, and regime-dependent occurrence on TRAIN. Check rate and burst sensitivity on VAL.	Materialize sparse binary events. Inspect and disclose repeated event windows and apply the active-window checks. Ratio-scale grading yields 13/10/3. The Q6 active-window check blocks quality selection.
Observability	Estimate observation rates, persistence, state-transition probabilities, run lengths, and reporting regimes on TRAIN. Select on VAL, then fit the final mask generator on TRAIN+VAL.	Generate reporting masks as target processes. Enforce their consistency before value scoring. Q3 supports limited diagnostic analysis. The branch is not retained in the Q6-selected subset.
Excluded / placeholders	Do not generate excluded roles. Assign placeholders through the role manifest.	Retain only where needed for namespace continuity. No positive realism, utility, coupling, or quality-selection claim.

S3 Artifact Hierarchy, Role Ownership, and Candidate Scope

Main manuscript Section 3.3 gives the role inventory and final ownership counts. Table S5 distinguishes the dataset versions used in the selection-policy and baseline comparisons.

Table S5: Artifact hierarchy and selection interpretation.

Artifact	Shape	Selection interpretation
Full scientific artifact	$255,540 \times 555$	Complete artifact for internal quality analysis. It is not approved for unrestricted public release.
Candidate under individual-driver checks	$255,540 \times 46$	Candidate under individual-driver filtering, containing timestamp, router, Zigbee, and 23 sparse-driver columns. It is used to compare selection policies.
Q6 selected candidate	$255,540 \times 23$	Conservative selected candidate containing only time, router, and Zigbee columns with documented warnings. All sparse-driver columns are blocked from quality selection and no formal privacy guarantee is claimed.
Dropped roles	532 columns	Excluded from quality selection due to quality, role, release, documentation, or active-window constraints, including 23 group-blocked sparse drivers and 3 fatal sparse drivers.

The main manuscript’s model-family table gives the reasons for including the continuous, binary, and sparse-event candidates. The branch rules above specify what is fitted on TRAIN and selected on VAL. Empirical context replay means reusing training observations under the branch’s conditioning context. The selected family is subject to the same TEST and release checks as the other candidates.

S4 Metric Definitions and Threshold Rationale

This section defines the metrics and explains the thresholds used in Q1–Q6. The thresholds determine which claims the study supports. They are policy choices rather than universal constants. The saved metric components allow alternative policies to be examined.

Main manuscript Section 3.5 defines the observation policy, KS distance, folded classifier AUC, and residential event–response scores. This section adds support overlap and threshold details. Section S10 gives the implementation-specific preprocessing and profile conventions.

The observed support of a feature is the set of distinct values present in the evaluated data. For example, a binary switch with both states observed has support $\{0, 1\}$. For low-cardinality features, let $\mathcal{S}_j^{\text{real}}$ and $\mathcal{S}_j^{\text{syn}}$ denote the real and synthetic supports. Their Jaccard overlap is

$$J_j^{\text{support}} = \frac{|\mathcal{S}_j^{\text{real}} \cap \mathcal{S}_j^{\text{syn}}|}{|\mathcal{S}_j^{\text{real}} \cup \mathcal{S}_j^{\text{syn}}|}.$$

The intersection $\cap$ contains shared values, the union $\cup$ contains values present in either set, and vertical bars count those values. For nonempty evaluated supports, 1 means identical sets and 0 means no shared values. For example, real support $\{0, 1\}$ and synthetic support $\{0\}$ give an overlap of $1/2$. This check detects a missing state even when its low frequency gives only a small KS difference.

Table S6: Threshold rationale and sensitivity. These values define the stated claim policy. They are not universal quality constants.

Gate	Rule and timing	Rationale and sensitivity
Continuous subsets	Candidate C2ST cutoffs $\tau = 0.60$ and 0.55 , selected using TRAIN/VAL evidence.	TEST AUC changes from 0.680907 to 0.619553 and 0.596994 . Some targets still fail the final audit.
Uniform generator audit	KS warning/fatal at $0.20/0.35$, and C2ST warning/fatal at $0.60/0.70$. Fixed before the audit.	Provides a common screen on compatible scopes. Compact continuous scopes fail for the pipeline and external models. Results for a 46-column candidate do not establish the utility of the final selected candidate.
Q4 coupling	Block if ETA/manifest similarity < 0.50 , lag error > 5 s, or response error > 0.25 . Policy specified using TRAIN/VAL.	Prevents a passing metric from overriding a failed required pair. Generic and all-eight claims fail. The later six-pair subset requires a fresh holdout.
Sparse-driver metrics	Rate, burst-count and windowed-Fano ratios. Inter-arrival and duration KS. Sparse-domain checks.	Ratio-scale P/W/F: $13/10/3$. Stricter Fano difference ($w = 0.1, f = 0.5$): $6/14/6$. Less strict Fano difference ($w = 0.5, f = 1.0$): $11/12/3$.
Q6 selection	Check role eligibility, documented warnings, manifests, and quality/active-window blockers.	Reduces 555 scientific columns to 23 candidate columns. All sparse drivers are excluded. Quality-selection eligibility is not inferred from Q5 utility.
Baseline comparison	Require compatible scopes and complete coverage. Use the recorded fixed TRAIN-only budgets.	The pipeline has lower penalties in seven pairwise comparisons across four complete feature scopes. External models receive no VAL selection, so these results do not establish superiority under equal tuning.

S4.1 Mapping to Classical Data and Information Quality Concepts

Table S7 relates the six dimensions to classical data and information quality concepts. The correspondence is not exact because CPS synthetic data also require checks of timing, observability, relationships between modalities, and quality-selection eligibility.

Table S7: Mapping Synthetic Fusion quality dimensions to classical data/information-quality concepts.

Synthetic Fusion dimension	Classical DQ/IQ concept	Application to CPS data
Q1 Marginal fidelity	Accuracy and representational consistency	Value, support, and distributional agreement between real and synthetic feature values.
Q2 Temporal coherence	Timeliness, consistency, and volatility	Transition, dwell, run-length, autocorrelation, and burst behavior.
Q3 Observability	Completeness and contextual completeness	Missingness, staleness, activity, and reporting behavior treated as semantic signals.
Q4 Cross-modal consistency	Semantic and inter-record consistency	Event-response timing, lag behavior, and protocol coupling between physical and network layers.
Q5 Task and baseline diagnostics	Fitness for use and task relevance	Comparisons with baselines, diagnostic tasks, and exploratory regression on the 23-column selected candidate that meets the numerical gap limits, each limited to compatible feature scopes.
Q6 Candidate selection	Subset documentation and selection governance	Candidate subset eligibility, warning governance, and role-level candidate-selection decisions.

S5 Artifact Package, Audit Procedure, and Temporal Stability

Table S8: Entry points in the separate empirical artifact package. These files are not included in the Overleaf source archive.

Check	Files and purpose
Start	README and package index identify the evidence files and table/figure links.
Integrity	<code>verify_package.py</code> checks the files against <code>MANIFEST.json</code> .
Fixed inputs	<code>check_manifest.py</code> verifies the supplied coupling manifest against the bundled TRAIN/VAL records. The manifest is a fixed data-derived input.
Schemas and values	Schema catalog, schema directory, table/figure maps, and paper-value ledger define fields and connect reported values to source rows.
Execution	<code>preflight.py</code> checks inputs and environment. The master notebook runs generation and evaluation. <code>compare_runs.py</code> compares two fresh runs.
Temporal stability	Window/block summaries and the uncertainty README report temporal dispersion, not independent-sample confidence intervals.
Smart* audit	In the Smart* manifest directory: <code>smartstar_gate_freeze_order_audit.csv</code> and <code>smartstar_gate_value_diff_against_primary_policy.csv</code> . These record artifact order, hashes, gate values, and scope declarations.
Protocol-activity Q5	Task settings, feature and sample lists, evaluation predictions, and score tables describe the regression in Section S10.
TON_IoT	<code>artifact_package_index_TONIOT_PHYSICAL_Q4_BRIDGE.csv</code> and checksum and package manifests connect the public Q4/Q6 study to its evidence.

Table S9: Evidence ledgers and selection-scope files used by the one-pass audit.

Group	Files	Claim audited
Role and branch QA	Role manifest and branch-readiness ledgers.	Structural validity, role ownership, and branch-readiness claims.
Q4 coupling	Q4 blocker and scope-status ledgers.	Generic Q4 blockers, manifest-subset blockers, all-8 A0 no-promotion, and 6-pair development evidence.
Sparse drivers	Sparse-driver ratio-scale audit ledger.	Final sparse-driver ratio-scale grading and quality-selection eligibility.
Release scope	Q6 selection manifest, selection-scope table, active-window diagnostics, policy-composition table, and conservative selected candidate directory.	Reported 23-column selected candidate, earlier 46-column candidate, and decisions to retain, exclude, or block columns.
Data access	Study repository and dataset documentation.	Code and residential data access are documented in the manuscript availability statement. Quality selection and permission to distribute remain separate.
Q5 consequence, candidate-scope regression, and baselines	Consequence-study ledgers, Q5 public-23 regression freeze/result tables, and baseline-fairness ledgers.	Diagnostic TRTR/TSTR consequence evidence, negative candidate-scope regression diagnostic, and fair-scope external-baseline comparison.
Smart* transfer study	Smart* role/split manifests, candidate-selection ledger, TEST-evidence ledger, gate-freeze audit, and gate-value diff files.	Procedural governance-transfer evidence. Confirms final-run record order and exact grade/membership joins. The gate-value diff is inconclusive and does not establish unchanged thresholds.
TON_IoT physical-Q4/Q6 transfer	TON_IoT bridge index, physical-Q4 TEST audit ledger, warning-pair table, Q4 scope summary, Q6 selection-scope table, and checksum manifest.	Second public transfer study over processed IoT telemetry and processed network-flow datasets. Q4 compares real TRAIN+VAL and real TEST profiles. TEST event frequencies were available during design.

Table S10: Pooled and window metrics on TEST. Windows describe variation within the trace. Logistic AUC and source-branch rank AUC use different calculations, detailed in Section S10.

Metric	Summary	Interpretation
Continuous KS, full TEST	Mean 0.379976 over 148 targets; see main manuscript Table 9.	Pooled summary across TEST. It does not quantify uncertainty across deployments.
Continuous KS, TEST windows	Mean 0.5032, SD 0.0309, range 0.4655–0.5439	Higher than pooled KS. The difference is descriptive. Matched sample sizes and classifier protocols are needed to isolate a temporal-pooling effect.
Continuous AUC, full TEST	Logistic AUC 0.766625. Source-branch rank AUC 0.680907	Logistic scores average held-out folds after full-sample scaling. The rank statistic directly compares values. They are different diagnostics.
Continuous C2ST/AUC, TEST windows	Mean 0.795783, SD 0.0118, range 0.7814–0.8123	Compare with the logistic pooled value 0.766625. The score is higher for these windows, but changed estimation conditions may contribute.
Router protocol KS, TEST windows	Mean 0.044, range 0.013–0.074	Describes protocol stability. It does not measure Q4 coupling.
Zigbee protocol KS, TEST windows	Mean 0.118	Describes protocol stability. The all-eight-pair Q4 claim remains blocked.
Sparse-driver KS, TEST windows	Mean 0.00036	Describes stability within windows. Ratio-scale metrics inform the scientific assessment, and the Q6 active-window check determines quality-selection eligibility.

The router, Zigbee, and driver window summaries use two contiguous halves of TEST. Their six rows are reproduced by the documented window-analysis implementation in the code package. The calculation is documented as a reconstruction because the original source was not recovered. The reported values are computed from the current generated outputs with this implementation.

S6 Extended Result Tables

The main manuscript reports the branch grades, Q4 pair-set summaries, policy comparisons, and task scores. The tables here add protocol coverage, exact selection diagnostics, continuous subsets, and baseline definitions.

Table S11: Protocol-side readiness and coverage in the final run.

Protocol scope	Retained evidence / status	Applicable dimensions	Selection / claim interpretation
Router Ethernet and Wi-Fi features	18 retained candidate columns. Subset retained with warnings.	Q1, Q2, Q5, Q6. Q4 only where event-response pairs are evaluated.	Restricted protocol diagnosis under the aggregate/count-like release rule. Downstream utility and formal privacy are not established.
Zigbee protocol features	4 retained candidate columns. 8 A_0 coupling candidates. All-8 Q4 blocked. 6-pair subset development-only.	Q1, Q2, Q4, Q6.	Supports coupling diagnostics for this deployment. A positive final Q4 claim is not established.
OTA wireless observations	2 scientific-scope columns. Not retained in $\hat{\mathbf{X}}^{\text{pub}}$.	Q1, Q2, Q6. Q4 only if paired.	Scientific/internal evidence only. No public-release claim.
Z-Wave indicators	0 columns retained in the clean logical scientific schema.	Q1, Q2, Q4 where paired, Q6.	No public-release or positive Q4 claim in this execution.
Generic all-protocol coupling	624 evaluated cross-modal pairs. 623 blockers.	Q4.	Blocked limitation evidence. No all-protocol coupling claim.
A_0 all-8 Zigbee candidate	8 evaluated pairs. 1 final blocker.	Q4.	Blocked. The all-eight-pair A_0 claim is not supported.
A_0 six-pair exploratory Zigbee subset	6 pairs. Zero current-TEST blockers. Mean ETA 0.710651.	Q4 diagnosis and development.	Development evidence only. Independent data are needed to confirm this pair scope.

Table S12: Logical columns in the Q6 selected candidate.

Retained group	Count	Role in selected candidate
Canonical timestamp/alignment column	1	Provides temporal alignment. Not separately scored as a CPS feature.
Router protocol columns	18	Restricted aggregate/count-like router columns retained with warnings. No established downstream utility.
Zigbee protocol columns	4	Zigbee protocol features retained under Q6. Q4 claims remain blocked or limited to development.
Sparse IoT event-driver columns	0	Blocked from the selected candidate by conservative Q6 active-window evidence. Ratio-scale evidence remains restricted scientific/selection-decision evidence.
Logical candidate columns	23	Candidate schema containing timestamp, router, and Zigbee columns after sparse drivers are blocked.
On-disk stored fields	23	The selected file is written without a persisted row index, so stored fields and logical columns have the same count.

Table S13: Q6 active-window check and separate real-split context over the 23 individual-grade sparse-driver columns. Exact sparse-window matches and zero-Hamming proximity are compared with one real-versus-real split and are not standalone blockers. The recorded group exclusion is driven by a difference in the fraction of windows with any event activity.

Diagnostic	Real-vs-real context	Synthetic vs real	Interpretation
Sparse-driver columns / windows	23 columns. 2,129 reference and 2,130 holdout windows	23 columns. 4,259 synthetic windows	One-second TEST rows are summarized in 60-row active-window vectors.
Fraction of windows with any event activity	Full real 0.266025. Real-split query 0.302347	Synthetic 0.621273. Full-real ratio 2.335393. Synthetic-minus-null 0.318925	Primary quality-selection blocker: the gate uses the full-real rate for both the ratio threshold 2.0 and the absolute-difference threshold 0.25. Both conditions hold. The real-split difference is a separate diagnostic.
Exact active windows seen in reference	0.934783	0.060469. Synthetic-minus-null -0.874314	Context only: exact sparse signatures also occur in this real-vs-real comparison, so nonzero exact matches do not by themselves prove training replay or justify exclusion.
Duplicate active-window member rate	0.961180	0.942555. Difference -0.018625	Context only: synthetic active-window repetition is high in absolute terms but not higher than the real-vs-real sparse-window null.
Repeat-after-first active-window rate	0.914596	0.895314. Difference -0.019283	Context only: this alternative repetition statistic is also not worse than the real-vs-real null.
Nearest-reference Hamming p01	0	0	Context only: zero-distance nearest neighbors occur under real-vs-real sparse-window self-similarity and are not standalone blockers.
Q6 checked decision	—	Block sparse-driver quality selection	Sparse drivers remain excluded from the selected candidate, but the blocker rationale is a difference in the fraction of windows with any event activity rather than uncalibrated exact-match or Hamming evidence.

Table S14: Continuous feature subsets. Membership is selected using VAL, and TEST results are reported after selection.

Subset	Targets	TEST AUC	TEST failures	Claim
Full branch	148	0.680907	27	No uniform realism claim.
$\tau = 0.60$	96	0.619553	14	Diagnostic subset. Not positive realism.
$\tau = 0.55$	71	0.596994	9	Diagnostic subset. Not positive realism.

Table S15: Exact component families used in baseline penalties. Each penalty is the arithmetic mean of its finite components. All 13 evaluated records contain five finite components.

Scope	Metric families included	Weighting and status
IoT compact tabular	Mean KS, histogram total variation, normalized quantile distance, correlation MAE, and excess folded AUC above 0.5.	Equal component weights. 52 columns. Support Jaccard is reported separately and is not in the penalty.
Protocol-core tabular	Same five tabular components.	12 columns. Equal component weights.
46-column candidate core	Same five tabular components.	45 evaluated fields from the 46-column candidate, including sparse drivers. Not the current 23-column scope.
Compact CPS sequence	Lag-1 error, mean multi-lag error, transition-rate error, run-length KS, and activity-rate error.	28 columns. Equal component weights. Marginal metrics are recorded separately.
Protocol sequence	Same five temporal components.	Pipeline coverage is 1 of 32 columns. This partial comparison is excluded from complete-scope counts.

Table S16: External-baseline scopes and fixed budgets. External baseline models fit on TRAIN, receive no VAL selection or calibration, and use TEST only for the compatible-scope audit.

Comparator	Compatible scope	Fixed budget
CTGAN	Compact IoT, protocol core, and 46-column candidate tables.	Batch size 500; 20 epochs; 20,000 TRAIN rows; CPU; timeout 7200 s per scope. Protocol counts are numeric in the latter two scopes; binary indicators remain categorical.
TabDDPM	Compact IoT, protocol core, and 46-column candidate tables.	Batch size 1024. 100 diffusion steps. 12 epochs. Hidden dimension 256. 100 sampling steps.
TimeGAN	Compact CPS sequences and the partial protocol-sequence scope.	Batch size 128. 8 autoencoder and 8 adversarial epochs. Hidden dimension 96. Sequence length 64. Stride 32.

The comparison ledgers record seeds, runtime, coverage, and failures/exclusions. The pipeline comparator in the partial temporal protocol row covers 1 of 32 columns and is not counted as a complete comparison.

S7 Deployment Transfer and Public-Data Details

Main manuscript Section 6 discusses validity and interpretation. This section gives the procedure for a new deployment and the detailed public-data records.

Table S17: Deployment-transfer protocol for applying Synthetic Fusion to a new CPS trace.

Component	Reusable?	How to apply
Role taxonomy	Reusable template	Map each new column to one owner role before generation. Add deployment-specific roles only through an explicit manifest extension.
Branch generators	Reusable classes, deployment-specific fits	Develop models on TRAIN and select on VAL. Record any final TRAIN+VAL fitting before TEST evaluation.
Q1–Q6 dimensions	Reusable	Retain dimensions but calibrate numerical thresholds to the new claim policy before TEST interpretation.
Q4 event-response manifest	Partly reusable	Define physical-event drivers and protocol responses from the new deployment. Freeze eligible pairs before TEST audit.
Q6 selection policy	Reusable logic, deployment-specific decisions	Apply role eligibility, blocker status, warning disclosure, and manifest completeness. Do not infer release from utility alone.
Artifact package	Reusable structure	Emit README, package index, schemas, checksums, table-to-ledger map, release manifest, restricted-trace statement, and one-pass audit script.

Table S18: TON_IoT application of Q4 and Q6. The study uses processed IoT telemetry and network flows. Pairs use TRAIN/VAL definitions. TEST event frequencies were available during design, so the comparison is exploratory.

Component	TON_IoT result
Dataset and source policy	Public TON_IoT processed IoT telemetry plus processed network-flow datasets. Train/test balanced extracts are not used as the primary Q4 source because chronological event-response auditing requires aligned temporal structure.
Aligned split	7,024 one-minute windows: 4,214 TRAIN, 1,404 VAL, and 1,406 TEST.
Role inventory	18 physical source columns, 40 derived physical drivers, and 30 Q4-eligible physical drivers after TRAIN/VAL assessment.
Q4 pair evaluation	50 pairs compare real TRAIN+VAL and real TEST profiles, giving 0 pass, 6 warnings, 40 metric failures, and 4 cases of insufficient TEST event support. The last two groups form the 44 blockers in the recorded audit.
Q6 projection	185 public-source aggregate columns kept with warnings and 1 split-control column excluded.
Claim consequence	Compares real TRAIN+VAL and real TEST profiles. Retained-row windows cross some clock gaps. Section S10 reports uninterrupted-minute sensitivity. This is not a synthetic-fidelity evaluation.

Table S19: Smart* Home A transfer study. This public trace tests the procedure using non-protocol roles. The primary deployment’s quality results are not assumed to generalize.

Component	Smart* evidence	Interpretation
Data	Home A archive: electrical, environmental, and operational signals.	No separate network stream, so protocol roles are absent.
Feature roles	Assigned/audited: continuous 64/64. Binary 33/31. Drivers 99/99. Observability 97/97.	2 binary features are excluded by TRAIN/VAL eligibility before TEST.
Splits	132,480 one-minute windows: 79,488 TRAIN, 26,496 VAL, and 26,496 TEST.	Final-run schema, roles, candidate registry, and Q1–Q3 gates are fixed before the reported TEST audit.
TEST results	P/W/F: continuous 5/10/49. Binary 27/2/2. Drivers 75/11/13. Observability 55/11/31.	Total audited scope: 291 eligible features. These outcomes restrict quality claims.
Q4	Not applicable: no independent network measurements.	No physical–network consistency claim.
Q5	Not assessed because no downstream prediction task is defined for this application.	No task-utility transfer claim.
Q6	196 keep-with-warning and 95 exclude decisions.	Tests selection rules on public-source data. It does not prove privacy.

Role counts precede eligibility filtering, while P/W/F counts follow it. The threshold records in Section S10 document the final run. They do not establish performance on a fresh holdout.

S8 Extended Design and Selection Tables

Table S20 links the research gap to the checks used by the method. Table S12 gives the candidate composition. Figure S1 shows the two selection stages.

Table S20: Design requirements and the checks they enable.

Requirement	Risk addressed	Method and evidence
Explicit owner roles	A feature can be handled by an unsuitable branch.	Assign one owner before fitting. Check the role manifest and ownership violations.
Observed-only values	Scores can include values when a device is unobserved.	Enforce active masks before scoring. Evaluate reporting patterns separately under Q3.
Separate dimensions	An aggregate score can hide a failure.	Record Q1–Q6 results for each applicable scope.
Defined coupling scope	A narrow relationship can be generalized to all protocols.	Freeze event/response pairs and apply pair-level blocker rules.
Compatible baselines	A comparison can mix unsupported roles or incomplete coverage.	Record feature scope, coverage, budget, and failure status.
Separate selection decision	Scientific use can be mistaken for quality-selection eligibility.	Apply Q6 role and blocker filters. Retain a documented selected subset.
Traceable evidence	Reported values can lack a checkable source.	Link claims and values to ledgers, schemas, and checksums.

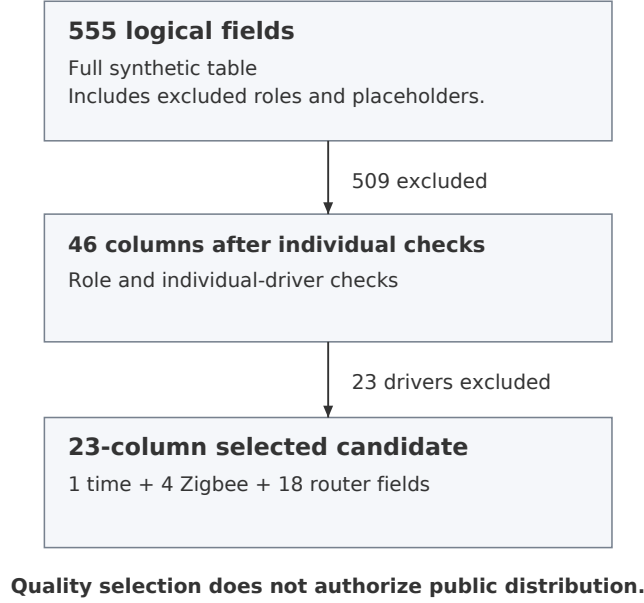


Figure S1: Q6 quality selection: 555 columns, 46 after individual-role and fatal-grade filtering, and 23 after the active-window check. Counts refer to logical features; the selected file has no additional stored index. This projection is not authorization to distribute data.

S9 Decision Rules and Controlled Evaluation

This section adds implementation checks for claim assessment, illustrates the treatment of observations and time, and describes the controlled experiment. Section S10 gives the calculations used for the empirical data.

S9.1 Claim Records and Evidence Status

Main manuscript Section 3.5 defines claim records, measurement records, and assessment states. The implementation additionally checks dataset, scope, and policy identities, including the compatibility of measurement protocols. File identity alone does not establish a compatible comparison. An empty requirement list cannot support a claim. All applicable failure and missing-evidence reasons are retained, even when one determines the reported decision.

S9.2 Observed Values and Clock Time

For a feature with real and synthetic observation masks M_t^r and M_t^s , let $n_r = \sum_t M_t^r$, $n_s = \sum_t M_t^s$, and $n_{\cap} = \sum_t M_t^r M_t^s$. The fractions $n_{\cap}/n_r$ and $n_{\cap}/n_s$ describe the overlap when their denominators are positive. Finite-value counts are recorded separately. Comparing values at shared observed times answers a different question from comparing each dataset's own available values. Neither comparison reconstructs values that were not observed.

For example, let both underlying sequences be $(0, 0, 10, 10)$, with real mask $(1, 0, 1, 0)$ and synthetic mask $(1, 1, 0, 0)$. Both observation rates are $1/2$, and the shared observed position contains zero in both sequences. However, the means over each sequence's own observations are 5 and 0. Equal observation rates therefore do not imply that value comparisons use equivalent populations.

A clock-time lag ℓ requires observations at timestamps t and $t + \ell$. Removing missing rows must not shorten the elapsed time. With observations only at seconds 0, 2, and 4, for example, no valid one-second pair exists. Run lengths also depend on whether a run reaches a gap or

Table S21: Observation rules that must be stated for a feature.

Situation	Interpretation	Treatment
Reported OFF/zero	A value is observed despite physical inactivity.	Preserve the value.
No report	No new observation is available.	State whether the entry is missing or carries an earlier value.
Stale reading	An earlier report is carried forward.	Record its age separately from its value.
Split boundary	The latest report may precede the split.	Use permitted past context or initialize age as unknown.

interval boundary. These cases should be distinguished from complete observed runs. Staleness resets when a report arrives and is unknown before the first available report unless earlier context is given. Section S10 states the retained-row conventions used in the empirical calculations.

S9.3 Result Scopes and Baseline Comparisons

Branch outputs and the assembled table can contain different values for the same name. The 148-name continuous source list, for example, includes the cups field whose assembled value comes from the binary branch. Each result must therefore identify both its feature list and the values being scored. Averaging a subset of stored per-feature scores is valid only when the values, preprocessing, model, and evaluated rows are unchanged. A change to any of these inputs requires the affected measurements to be computed again. Section S10 reports the 147-field aggregate and distinguishes source-value from assembled-value summaries.

Table S22: Recorded baseline penalties on the stated feature scopes. Lower is better within a row. These fixed-budget diagnostics do not establish model superiority.

Scope	External models in scope	Pipeline penalty	Best external penalty
<i>Complete comparisons on compatible scopes</i>			
Compact IoT tabular	CTGAN (best), TabDDPM	0.354782	0.361808
Protocol-core tabular	CTGAN, TabDDPM (best)	0.074814	0.150078
46-column candidate table	CTGAN, TabDDPM (best)	0.062348	0.201483
Compact CPS sequences	TimeGAN	0.048831	0.456915
<i>Partial comparison, excluded from the complete count</i>			
Protocol sequences (pipeline: 1/32)	TimeGAN	0.062176	0.328602

Penalties use five equally weighted components within each metric family (Section S10). Values should not be ranked across rows. The 46-column candidate row concerns the 46-column candidate, not the final 23-column candidate. External models use TRAIN-only budgets without VAL selection or calibration. The best external penalty is the lowest recorded penalty among the external models in that scope. Tables S15 and S16 give the metric families and budgets.

S9.4 Event–Response Interpretation

Q4 compares response profiles, peak timing, and response magnitude for specified event–response pairs. The event rule, window, transformation, normalization, support requirement, and constant-profile convention all affect the result. Residential A_0 and TON-IoT use different definitions, which are given in Section S10. TON-IoT compares two real periods, whereas A_0 compares real responses with an empirical construction at synthetic event times.

The ETA equation gives 1.000000, 0.524894, and 0.500023 when correlation is 1 and normalized error is 0, 3, and 10, respectively. A perfectly correlated profile can therefore remain above 0.5 despite a large magnitude error. The additional response-error and profile checks are needed to apply the complete Q4 policy. Constant zero profiles require separate interpretation because a numerical similarity does not establish a resolved physical response.

S9.5 Sparse Activity and Task Scores

Sparse metrics depend on the event definition: a positive row, rising edge, and complete episode are different counting units. A ratio also requires a defined denominator. Zero real rate with positive synthetic rate indicates a mismatch, while two zero rates give an undefined ratio rather than evidence of preserved nonzero activity. Inter-arrival and duration statistics require enough events or episodes for their respective samples.

The active-window rate is $u = N^{-1} \sum_w \mathbf{1}\{\text{at least one retained driver is active in } w\}$. The Q6 rule uses $u_s/u_r \geq 2.0$ or $|u_s - u_r| \geq 0.25$, with the full-real comparator and $N = 4,259$ one-minute windows. The observed difference is $0.621273 - 0.266025 = 0.355248$. The relative term targets excess activity. For example, $u_r = 0.10$ and $u_s = 0$ would pass both components, although other event checks could still fail. The thresholds therefore express this study’s policy rather than a universal relative error tolerance.

The repeated-pattern comparison uses 23-bit presence vectors. The first 2,129 real windows form a reference and the remaining 2,130 a query set. This comparison loses within-minute timing and does not control differences between time periods. It provides context for sparse-pattern repetition, not a test of training-data memorization or calibration of the separate active-window threshold.

Task scores also need a clear direction. For common evaluation cases, define classification loss as $L_{\text{AUC}} = \text{AUC}_{\text{TRTR}} - \text{AUC}_{\text{TSTR}}$ and error loss as $L_{\text{err}} = \text{MAE}_{\text{TSTR}} - \text{MAE}_{\text{TRTR}}$. Positive values indicate worse performance after synthetic training. Absolute gaps hide this direction, so both model scores should accompany them. Real-derived model inputs and target transformations must also be stated, as in the protocol-activity regression.

Table S23: Absolute classification scores for seed 20260702. Signed losses use unrounded scores. These results describe one learner seed.

Task / scope	TRTR AUC	TSTR AUC	Signed loss
T1 / R1	0.893650	0.806258	+0.087391
T1 / R2	0.888628	0.761030	+0.127598
T1 / R3	0.878702	0.832958	+0.045744
T2 / R1	0.725051	0.475963	+0.249088
T2 / R2	0.727007	0.524535	+0.202472
T2 / R3	0.871406	0.485355	+0.386051

All reported T1 and T2 signed losses are positive in this seed. Absolute gaps should still be read with both model scores because a small gap does not establish good predictive performance. Section S10 gives the diagnostic task definitions, fitting-size differences, and normalization choices. These conditions limit comparisons between feature scopes.

S9.6 Executable Controlled Experiment

The controlled experiment provides executable Python code, a notebook, the fixed protocol, trial-level results, summaries, and environment records. The calculation requires NumPy and does not use private data or train neural generators. Pilot seed 319407 and final seed 928361 provide separate random sequences. Reference and candidate samples are also generated independently within each trial.

There are 100 trials per candidate parameter at each scope, 32 pilot pairs, and 400 low-support pairs. All four rules use the same samples, probability tolerance 0.01, and minimum sample size 128. The policy and trial count were set before the pilot. The pilot assessed runtime and did not tune the thresholds. The sample minimum is an illustrative support rule, not a precision guarantee.

Main manuscript Section 5.1 reports the trial outcomes and acceptable-case coverage. The supplementary implementation checks compare the decision code with an independently written Boolean specification and analytical examples. These checks concern the stated binary setting and decision logic, rather than calibration of the residential C2ST or Q4 measurements.

S10 Implementation Details and Additional Analyses

This section specifies the calculations underlying the residential and public-data results. It distinguishes source-branch outputs, assembled values, and selected feature lists so that each result has a clear scope.

S10.1 Feature Inspection and Role Interpretation

The feature inspection covers the 115 flagged fields on the canonical input’s TRAIN interval, rows 0–766,615. The input contains 740 columns and 1,277,694 rows. Its TRAIN timestamps follow a complete one-second grid. This input table is distinct from the assembled 555-column synthetic table. Inspection with seed 20260907 samples up to ten nonmissing values and ten distinct values per field. The supporting records contain 2,992 sampled values, 1,620 temporal-context records, and 195 distinct aligned example rows. Context windows extend at most two rows from each selected example. These samples illustrate feature behavior rather than independent events.

Main manuscript Section 3.3 reports the observed-value categories, author-assigned meanings, and the milk-cup assembly exception. The classification-to-generation cross-check adds the following detail: 74 of the 94 continuous-value labels match continuous generation roles, while 20 concern excluded fields. Ten of the 19 binary labels match binary generation roles, while nine concern excluded fields. Thus all 29 fields without TRAIN observations remain excluded regardless of their semantic label. Among the 13 constant fields, six have binary and seven continuous generation roles. The observability label is `iot__tier_present`.

Temporal examples help narrow the possible interpretations. The aggregate fields `iot__events_total` and `iot__events_entity_unique` both contain $\{0, 1, 2, 3\}$, with 9,925 and 9,922 decreases between adjacent finite values. They are not monotone cumulative series on TRAIN. Door/window sensor PID005’s trigger-count field contains $\{0, 1, 2\}$, with 24 decreases and a longest positive run of 15,625 rows. PID007’s field contains $\{0, 2, 4, 6\}$, with 12 decreases and a longest positive run of 9,453 rows. These decreases do not by themselves identify device resets. Similarly, light PID002’s brightness values $\{146, 255\}$ describe two observed brightness levels rather than a Boolean state.

Reporting indicators also differ. Among the 195 selected example rows, 121 have `iot__tier_present=1` and `iot__obs_present=0`. This shows that the stored indicators are distinct, without estimating how often the condition occurs in the full trace. The indicators `iot__events_update_any` and `iot__any_update_raw` differ on 154,652 TRAIN rows, giving a

match fraction of 0.798267. They cannot be substituted for one another without changing the data. Upstream extraction definitions are needed to establish their precise meanings.

S10.2 Continuous Scores and Observation Populations

The source-branch rank AUC and the assembled-data logistic AUC use different calculations. Both remove nonfinite values separately from real and synthetic data. They therefore compare each dataset’s available values, rather than values restricted to shared observed times.

For rank AUC, each feature contributes at most 4,096 values per class, sampled without replacement with seed 123. The feature values themselves are ranked, and their rank-sum AUC is folded as $\max(a, 1 - a)$. No classifier is trained. The implementation returns 0.5 if either class has fewer than 32 finite values. That fallback cannot establish similarity, but it is not activated in the 148 recorded comparisons because each has finite-value rate 1 in both datasets. Finite values can include values filled upstream and do not necessarily indicate fresh observations.

The logistic evaluation fits a one-feature classifier with maximum 500 iterations. Real and synthetic values at equal retained-row positions share a group identifier. Four-fold GroupKFold provides held-out scoring groups. Standard scaling is fitted to the entire stacked comparison sample before the folds are formed, so evaluation observations influence preprocessing. AUC is folded within each fold and then averaged, with equal weight for each feature.

Each class must have at least 100 nonmissing values. The pooled calculation uses up to its first 20,000 retained values, while each of six original-row windows uses up to 10,000. Four evaluation folds are formed within each comparison, without a separating time gap. Because missing values are removed independently, matching retained positions need not represent matching clock intervals. The pooled and window scores also have different sample caps. Their means, 0.766625 and 0.795783, are therefore descriptive separability summaries rather than calibrated tests of equality or an isolated comparison of temporal scales.

The source continuous list contains 148 names and overlaps the binary list on `iot__coffee_maker_milk__sensor__cups__value`. The assembled table uses the binary output for this field, giving 147 continuous and 52 binary fields. Logistic scoring on the 148 source names includes the assembled cups score of 0.5. Removing that field gives a 147-field mean of 0.768439. On the source-branch values, the same 147-name subset has rank AUC 0.682137, KS 0.382561, and normalized Wasserstein distance 0.372560. These source-value summaries should not be read as measurements of the assembled continuous values. Window records contain only 148-name means, so a 147-field window mean is unavailable.

The full continuous branch has grades 70/30/48. Its restricted evaluation list excludes 29 fields comprising 4 passes, 4 warnings, and 21 fatals, leaving 66/26/27. The binary list excludes 4 passes, 3 warnings, and 6 fatals from 52 fields, leaving 27/9/3 over 39 fields. These are changes in scope. Excluded results do not become passes.

S10.3 Observation Masks and Assembly Checks

The observation-mask branch chooses indicator mode from field-name rules or binary TRAIN/VAL values. Other fields use raw-value presence mode. Indicator mode parses numeric or string states, thresholds at 0.5, and maps missing or unparseable entries to zero. Raw-value mode checks whether a value is present, so a recorded numerical zero remains observed. The modes have different meanings and depend on the field’s role. Run lengths use adjacent array positions and include runs that reach an array boundary.

After TRAIN-to-VAL model selection, the final mask generator is fitted to combined TRAIN+VAL states. The continuous assembly check covers 148 targets and 37,819,920 entries. It finds zero inactive mask cells, zero values changed to missing, and zero residual finite values at inactive positions. Since all checked mask entries are active, this result does not test behavior under missingness. It also does not count transformations in other branches.

Q3 separately evaluates 183 masks using rate, run-length, dwell-distance, and mask-AUC criteria. All use indicator parsing. The full list contains 59 passes, 62 warnings, and 62 fatals. A list defined on TRAIN/VAL excludes 90 fields: 29 passes, 5 warnings, and 56 fatals. The 93 included masks therefore contain 30 passes, 57 warnings, and 6 fatals. The six included fatals block their affected claims. The two summaries describe different feature populations under the same measured results.

S10.4 Diagnostic Prediction Tasks and Learner Variation

The diagnostic prediction tasks use 300-row histories, a 60-row prediction horizon, and stride 60 on the residential one-second grid. Each feature contributes its history mean, history SD, and final-60-row mean. T1 labels whether any of the same 23 drivers is positive in the future window. T2 uses the future mean of the same 18-router-field row sum, compared with its median on real TRAIN. Label definitions and real evaluation intervals are common across R1–R3. T3 predicts future-window means for the first ten continuous source names in lexicographic order.

Real fitting uses 766,616 rows and synthetic fitting uses 255,540 rows. R1 includes the canonical timestamp as an input. Each T3 MAE is divided by the error of its own fitting-target-mean predictor, so the real and synthetic fits have different denominators. These conditions limit an interpretation of gap differences as effects of feature scope alone. The five seeds vary the gradient-boosting learner on fixed data. In the recorded outputs, TSTR scores are identical across seeds within each task and scope, so the gap variation comes from TRTR.

S10.5 Protocol-Activity Regression

The regression uses the real TEST table and the 23-column synthetic candidate on the same one-second grid. The timestamp is excluded. Eight router and four Zigbee fields have positive IQR on the real fitting interval. The router fields are packet total, byte total, TCP ACK, TCP packet, UDP packet, ARP packet, ARP request, and ARP reply. The Zigbee fields are packet total, byte total, ACK, and command. Ten router fields have zero IQR and are excluded: TCP SYN/RST/FIN, ICMP, DNS packet/query/response, DNS response-code 0/3, and DHCP.

The real fitting interval is rows 0–102,179. Its medians and IQRs normalize both datasets. Normalized missing values are set to zero, the real median on this scale. Across the selected fields, router missing rates are about 0.155% in real data and 0.167% in synthetic data. Zigbee rates are about 11.615% and 8.124%. Median filling means that this task cannot separate reporting differences from activity differences. Q3 evaluates reporting behavior independently.

The target is next-minute mean absolute normalized activity over the 12 fields. Each current-minute window provides mean, population SD, minimum, maximum, and last value, giving 60 inputs. Of 4,258 possible current/next minute samples, 1,702 are used for fitting, one crossing the boundary is excluded, and 2,555 are used for evaluation. Fitting targets end at row 102,180 (exclusive), where evaluation inputs begin. Both models use real-data preprocessing. Five paired random forest fits use 300 trees, maximum depth 6, minimum leaf size 5, and seeds 20260710–20260714.

The real fitting-target population SD, 56.652090, normalizes MAE. Mean MAE is 21.930489 for TRTR and 27.148402 for TSTR, with normalized values 0.387108 and 0.479213. Signed normalized loss is positive in every seed, with mean 0.092105 and sample SD 0.009649. The median absolute gap 0.092861 and 95th-percentile gap 0.102925 meet the 0.10 and 0.20 limits. The 95th percentile summarizes five learner results and is not an estimate of rare deployment failures.

Persistence has MAE 12.087604, while fitting-target median and mean predictors have MAE 13.998192 and 25.244190. Their normalized errors are 0.213366, 0.247090, and 0.445600. Both random forest models have higher MAE than persistence and the median predictor. Mean R^2 is 0.217561 for TRTR and 0.049053 for TSTR, compared with -0.190853 for persistence. MAE

and squared-error criteria can rank models differently. MAE is the declared task loss. All scores agree with the 2,555 evaluation predictions per model. Since the task was developed using an inspected TEST period, the result remains exploratory.

S10.6 Residential Event–Response Evaluation

The A_0 diagnostic builds complete event windows within the data boundaries. Its configured protocol transformation defaults to signed $\log(1 + |x|)$. Each profile is centered by its mean pre-event value, or its first finite entry when no pre-event mean is available. For finite paired profile entries, normalized error and similarity are

$$E = \frac{\text{mean } |r - s|}{\text{mean } |r| + \text{sd}(r) + \epsilon}, \quad S = \frac{1}{2} \max(0, \rho) + \frac{1}{2} e^{-E}.$$

The default ϵ is 10^{-9} . Constant profiles receive correlation 1 if numerically equal and 0 otherwise. Peak lag is the first offset maximizing absolute profile magnitude. The default profile covers ten rows before and thirty after the event.

Response rate is the fraction of event windows crossing a configured quantile of absolute transformed real TEST values, with default quantile 0.90. Its error is therefore a dimensionless proportion. This threshold is part of the TEST statistic, not a threshold fitted on TRAIN. Runtime configuration determines whether these defaults are used. The Q4 implementations have different support and constant-profile conventions, which must be considered when interpreting grades across settings.

The eight-pair A_0 evaluation gives one pass, six warnings, and one fatal, with mean ETA 0.652368, lag error 0.25 seconds, and response-rate error 0.071313. Every pair has at least 92 real and 100 synthetic complete event windows. Recomputing similarity, correlation, MAE, and normalized MAE from the float32 profiles agrees with all 64 stored values within 1.38×10^{-7} .

The construction learns profiles on TRAIN/VAL and places responses around synthetic driver events. It changes 45,344 protocol entries, with no reported changes outside the target fields or event windows. The construction weight is 0.85 times the recommended weight. The archived in-memory base protocol has no recorded file hash. The reproduction workflow saves the base matrix before materialisation, allowing its identity to be checked in new runs. These constructed relationships are distinct from coupling learned by the A_2 branches.

The generic evaluation covers 624 pairs. Every pair meets the five-event minimum in each dataset, with minima of six real and nine synthetic events. ETA yields 9 passes, 20 warnings, and 595 fatals. Lag/response yields 3 warnings and 621 fatals, while cross-correlation yields 56 warnings and 568 fatals. Their union gives one warning and 623 blocked pairs. The blocker file also contains three group summaries and one overall summary, so its 627 rows do not represent 627 failed pairs. The pair list does not independently label expected responses and meaningful null relationships. A sufficient event count alone does not establish reference stability or a physical reason for each pair to respond.

S10.7 Baseline Penalties

The baseline evaluation contains 13 evaluable outputs and 416 field-level records. The two count-aware CTGAN outputs add 57 evaluated fields. Seven pairwise comparisons have complete coverage across four feature scopes. The eighth comparison is partial: the protocol-sequence pipeline comparator covers one of 32 fields, while TimeGAN covers all 32. Table S22 reports the lowest external penalty in each scope. The partial comparison is excluded from complete-scope conclusions.

For protocol-core and intermediate-candidate CTGAN, numeric counts are modelled as numeric inputs rather than categorical labels. The 23 binary event indicators in the latter scope remain categorical. These runs use the same 20,000 TRAIN rows, seeds, architecture and 20-epoch

CPU budget, with TRAIN-derived clipping and integer rounding unchanged. They generate 255,540 rows with 12 and 45 fields in 118.25 and 236.70 seconds, respectively. The compact IoT CTGAN configuration is unchanged. Configuration identities and the earlier budget-limited attempts remain in the computational record. Table S24 gives the component scores.

Table S24: Tabular baseline metrics within the two evaluated scopes. CTGAN uses numeric protocol counts and categorical binary indicators. Lower is better for KS, TV, normalized Wasserstein, correlation error and penalty; folded AUC closer to 0.5 indicates less separation.

Model	KS	TV	W_{norm}	E_{corr}	AUC	Penalty
<i>Protocol core: 12 fields</i>						
Pipeline	0.0764	0.0514	0.0683	0.0129	0.6651	0.0748
CTGAN	0.1970	0.0928	0.2865	0.1662	0.6887	0.1862
TabDDPM	0.1815	0.0672	0.2928	0.0599	0.6490	0.1501
<i>Intermediate candidate core: 45 fields</i>						
Pipeline	0.0275	0.0173	0.0777	0.0219	0.6674	0.0623
CTGAN	0.0797	0.0313	0.5734	0.0870	0.8103	0.2164
TabDDPM	0.0708	0.0255	0.5485	0.0694	0.7933	0.2015

Let M be the arithmetic mean of finite terms. The penalties are

$$P_{\text{tab}} = M(\overline{\text{KS}}, \overline{\text{TV}}, \overline{W_{\text{norm}}}, E_{\text{corr}}, \max(0, A - 0.5)),$$

$$P_{\text{time}} = M(E_1, \overline{E_{1,5,10,30,60}}, E_{\text{transition}}, \text{KS}_{\text{run}}, E_{\text{activity}}).$$

Each evaluated output has five finite components, weighted equally. Field averages can omit nonfinite values, whose counts are recorded. The tabular Wasserstein approximation averages absolute differences at 101 equally spaced empirical quantiles, divided by real-value population SD plus mean absolute real value and 10^{-9} . Histogram total variation uses the common real/synthetic range. Support Jaccard is recorded separately.

Correlation MAE averages entries that are finite in both the real and synthetic within-dataset Pearson correlation matrices, including their diagonals. Temporal penalties omit the separately reported marginal metrics. Lag-one error contributes both directly and within the multi-lag mean. Equal weights therefore do not imply independent or equally variable components. Temporal calculations independently remove nonfinite values, so their retained-row lags need not equal clock-time lags.

Baseline C2ST uses earlier retained positions for fitting and later positions for evaluation. Median imputation is fitted on the real earlier block, and scaling is fitted within the logistic pipeline. This differs from the residential per-feature logistic calculation. If fitting fails, the baseline implementation can return a univariate rank statistic. The current evaluation records whether this fallback is used. It was not used for either count-aware CTGAN output or the four re-evaluated pipeline and TabDDPM controls. All 1,172 aggregate and per-field control checks match their recorded values within relative tolerance 10^{-7} and absolute tolerance 10^{-9} . Fallback use remains unrecorded for the other original outputs, so their classifier components retain that limitation.

For protocol-core data, C2ST is lower for TabDDPM (0.648966) than for the pipeline (0.665109), despite the pipeline’s lower aggregate penalty. The 46-column candidate contributes 45 evaluated fields to its tabular comparison. Fixed external budgets, unequal access to VAL selection, and incomplete coverage prevent a general ranking of generator families.

S10.8 Smart* Evaluation

Smart* fills structural zeros in event and reporting indicators before determining feature eligibility. The evaluated list has 291 features: 162 pass, 34 warning, and 95 fatal. Q6 retains pass and warning features under one retention label and excludes fatal features, giving 196 retained and 95 excluded. The feature-level quality grades remain available.

Its C2ST uses gradient boosting with a stratified random split, rather than the residential logistic grouping. Continuous KS and normalized Wasserstein warning/fatal limits are 0.20/0.35, with autocorrelation-error limits 0.10/0.25. These define the Smart* application. Numeric matches between thresholds do not establish identical evaluation semantics across datasets. The recorded timestamps span 30 April–31 July 2012.

S10.9 TON-IoT Profiles and Clock-Window Sensitivity

TON-IoT alignment retains minute timestamps present in both telemetry and network tables, followed by a chronological 60/20/20 split. Physical-event thresholds use TRAIN+VAL. Candidate pairs are ranked on TRAIN/VAL profiles and the top 50 retained. TEST event frequencies were available during study design, so this is an exploratory analysis.

Q4 compares real TRAIN+VAL and real TEST profiles. Responses are scaled using TRAIN+VAL medians and IQRs, with SD and then unit scale as fallbacks. Profiles cover retained-row offsets $-5, \dots, 5$ without pre-event baseline subtraction. Similarity combines profile correlation with MAE divided by mean absolute reference profile, using unit scale when this denominator is near zero. Peak lag is the first maximum at offsets $0, \dots, 5$. Response error compares post-event profile means with a 10^{-9} denominator floor. These definitions differ from residential Q4.

The retained-row calculation gives 0 passes, 6 warnings, and 44 blockers. Four blockers have fewer than five TEST events. Forty have enough events but fail the metric gate. Five warning pairs (032–036) have identical zero profiles. Their zero correlation and zero normalized error give $S = 0.5$, with zero lag and response errors, placing them at the warning boundary. They provide no evidence of a resolved response. Pair 021 is the only warning with nonconstant profiles.

TRAIN, VAL, and TEST contain 29, 5, and 3 gaps, respectively, corresponding to 30,685, 1,304, and 168 absent minute bins. The largest TRAIN gap contains 27,619 absent bins. We assess the effect of these gaps by retaining the same 50 pairs and thresholds but using only event windows with eleven consecutive one-minute timestamps. Event definitions and TRAIN+VAL scaling are unchanged. This isolates window continuity within the selected pairs, without reselecting pairs or interpolating missing observations.

All 50 pairs lose at least one reference window, and 43 lose at least one TEST window. Pair 021 changes from warning to blocker, and pair 024 changes from blocker to warning. Other adequately supported pairs retain their grade. Four pairs remain insufficiently supported, and five warnings still have zero profiles. The total counts therefore hide a change in the identity of the only nonconstant-profile warning.

Table S25: TON-IoT results using the same pairs and thresholds with two window definitions.

Category	Retained-row windows	Uninterrupted minute windows
Pass	0	0
Warning with nonconstant profiles	1 (pair 021)	1 (pair 024)
Warning with identical zero profiles	5	5
Metric blocker with sufficient event support	40	40
Insufficient TEST event support	4	4
Total fixed pairs	50	50

Requiring $S > 0.5$ rather than allowing the boundary would remove the five zero-profile warnings from that component. This is a threshold sensitivity, not a validated alternative policy. The selected TRAIN/VAL list contains one pass, 36 warnings, and 13 blockers because ranking does not require an acceptable grade. Pair selection alone therefore does not establish a stable reference relationship.

Q6 retains 185 columns from the aligned real aggregate and removes the split indicator. This is column selection on real data, not a synthetic-data release. No synthetic dataset enters this calculation. For the residential synthetic table, the corresponding accounting is 23 retained and 532 excluded columns. Each selection result concerns its own dataset and quality policy.